

We present a Coronal Mass Ejection (CME) that impacted Earth from 2015 December 19th 16:30 UTC to 2015 December 21st 21:24 UTC. The event consisted of a forward shock, a turbulent sheath region, and a well-defined magnetic obstacle (MO). The MO contained a structured flux rope with two distinct southward B_z pulses, separated by a partial B_z recovery, followed by a final B_y -dominated interval. This four-phase internal structure of the MO provided a controlled natural experiment for isolating different regimes of solar wind–magnetosphere–ionosphere coupling. Through a multi-source data analysis we try to resolve the temporal and spatial evolution of the coupled system.

DATA

- ACE [Stone+, 1998] & Wind [Harten & Clark, 1995]; Solar wind IMF, speed, density; dual time-shifted to bow shock (+61 / +43 min) determined by cross-correlation with MMS [Burch+, 2016]
- OMNI [King & Papitashvili, 2005]; Sym-H, ASY-H
- PC-North polar cap index [Troshichev+, 1979]
- AMPERE [Anderson+, 2000 & 2008; Waters+, 2001]; Field-aligned current keograms at four MLT meridian pairs
- THEMIS-A [Angelopoulos, 2008]; Earthward energy flux
- SuperMAG [Gjerloev, 2012]; SMR (MLT-resolved ring current), SMU/SML electrojet indices
- IMAGE chain [Tanskanen, 2009]; Equivalent ionospheric currents (EIC) keogram

DISCUSSION

AMPERE observations show that IMF orientation changes are reflected not only in the strength of the field-aligned current system, but also in its morphology and local-time distribution.

The first southward- B_z interval produces a strong and well-organized R1/R2 response. The enhanced response during the second southward- B_z interval with comparable IMF magnitude raises the question whether the magnetosphere was preconditioned by the first interval. Entering the second pulse with an already-expanded polar cap and elevated open flux FPC may amplify the ionospheric response to a given Φ_D^* [Milan+, 2021].

The subsequent B_y -dominated phase is associated with a weaker but distinctly asymmetric FAC pattern, with more organized R1/R2 structure in the 15–21 MLT sectors than in the 03–09 MLT sectors. While Sym-H is recovering, ASY-H remains substantial and a significant longitudinal asymmetry persists. This persistence and the dawn-weighted ring current through the recovery phase suggests that injection continued beyond the main phase driving, possibly sustained by repeated tail-reconnection events. It might be that the observed BBF signatures are sufficient to account for this continued injection.

OPEN QUESTIONS

- Does an already-expanded polar cap and elevated open flux at the onset of the second southward B_z pulse amplify the ionospheric FAC response beyond what Φ_D^* predicts?
- What sustains the ring current asymmetry (ASY-H) into the recovery phase after Sym-H begins to recover, continued tail injection, suppressed loss, or both?
- Is the local-time asymmetry of the R1/R2 system during the B_y -dominated phase a direct imprint of IMF B_y , or does it reflect the residual storm-time state of the magnetosphere?
- Are the observed BBF energy flux signatures energetically sufficient to account for the continued ring current injection during the recovery phase?

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MULTI-POINT OBSERVATIONS OF A GLOBAL MAGNETOSPHERIC RESPONSE

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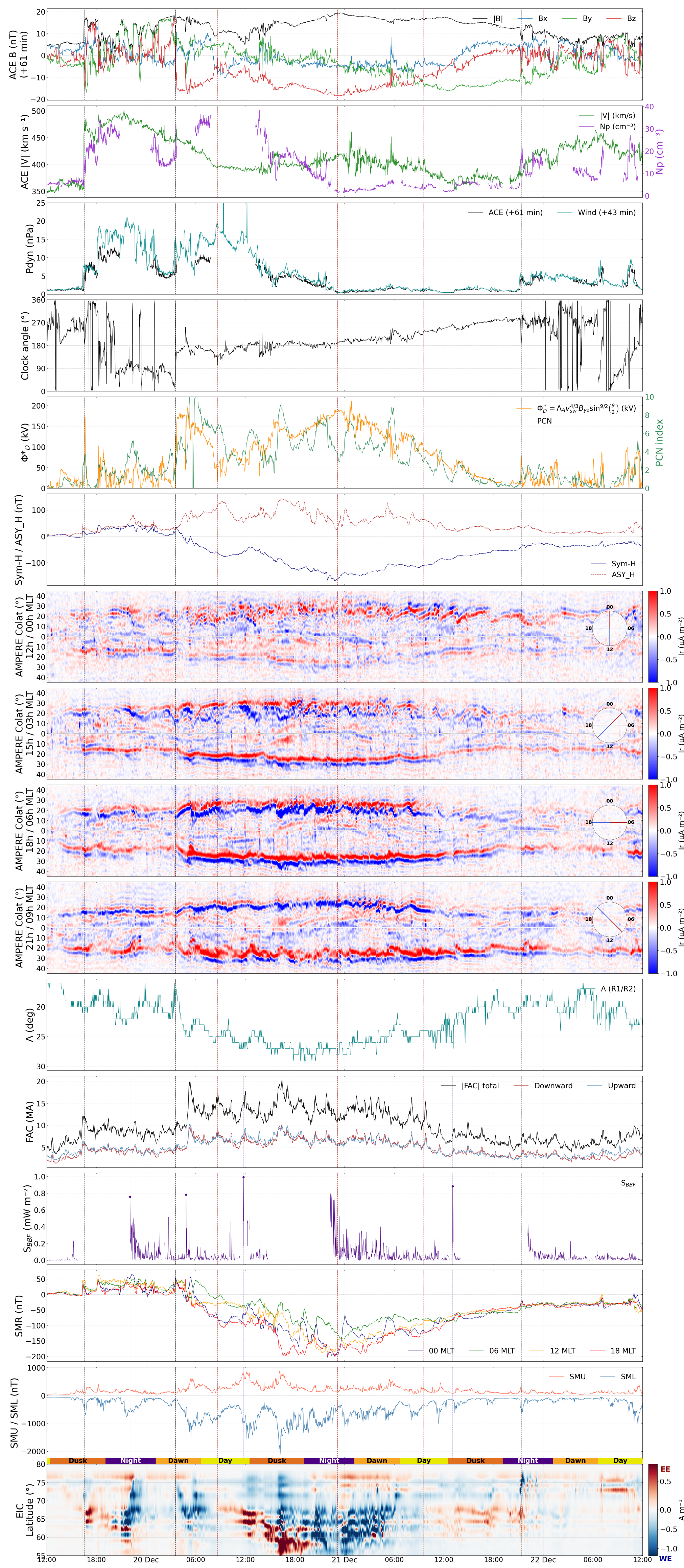


Figure 1. Multi-source overview of the December 2015 CME event. **Rows 1–2:** ACE IMF components and solar wind bulk parameters, time-shifted +61 min to bow shock arrival. **Row 3:** Dynamic pressure from ACE and Wind (+43 min). **Row 4:** IMF clock angle θ . **Row 5:** Dayside reconnection rate proxy Φ_D^* and polar cap index PCN. **Row 6:** Sym-H (ring current) and ASY-H (ring current asymmetry). **Rows 7–10:** AMPERE field-aligned current keograms at four MLT meridian pairs (00/12, 15/03, 18/06, 21/09 h MLT); red = downward, blue = upward FAC. **Row 11:** Λ -colatitude [Milan+, 2015] of the R1/R2 boundary; proxy for open magnetic flux F_{PC} . **Row 12:** Integrated hemispheric |FAC| with downward/upward contributions. **Row 13:** THEMIS-A earthward energy flux S_{BBF} ; arrows mark probable reconnection events. **Rows 14–15:** SuperMAG SMR (four MLT sectors) and SMU/SML electrojet indices. **Row 16:** IMAGE equivalent ionospheric current keogram. Vertical lines mark CME phase boundaries.

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